

Quantum Learning Platform

Complete Functional Breakdown & Short Notes Reference Guide

1. Landing Page & Testbench (/)

Primary Purpose: Welcomes visitors and provides immediate hands-on quantum interaction without requiring login or prior setup.

Key Features & Functions:

- **Live 3D Bloch Sphere Demo:** Enables visitors to apply single-qubit gates (H, X, Y, Z) and watch real-time 3D statevector rotations on the sphere.
- **Mathematical Readout:** Computes exact probability amplitudes, measuring odds $P(|0\rangle)$ & $P(|1\rangle)$, and quantum state equations.
- **Hero Search Bar:** Instant keyword search for quantum concepts, algorithms, and gate definitions from the platform catalog.
- **Interactive Walkthrough:** Summarizes platform stages (Learn, Experiment, Explore, Practice) with direct jump links.

2. Quantum Basics (/basics)

Primary Purpose: Teaches fundamental quantum concepts progressively using beginner-friendly expositions and interactive visual widgets.

Key Features & Functions:

- **7 Sequential Learning Modules:** Classical Bits vs. Qubits, What is a Qubit?, Computational States $|0\rangle$ & $|1\rangle$, Superposition, Probability Amplitudes, Measurement & Wave Collapse, and The Bloch Sphere.
- **Interactive Manipulatives:** Includes live probability amplitude ratio sliders, physical qubit diagrams, and polar (θ) / azimuthal (ϕ) Bloch controls.
- **Scientific Accuracy:** Strictly eliminates classical misconceptions (e.g. clearly frames superposition via amplitudes rather than 'both states at once').
- **Progress Flow:** Step-by-step previous/next navigation with persistent module completion tracking.

3. Quantum Lab (Circuit Studio) (/lab)

Primary Purpose: Full-featured drag-and-drop quantum circuit builder and simulation studio for designing and testing custom circuits.

Key Features & Functions:

- **Drag-and-Drop Gate Grid:** Multi-qubit circuit grid (up to 4 qubits) supporting single-qubit gates (H, X, Y, Z, S, T, RX, RY, RZ), two-qubit gates (CNOT, CZ, SWAP), and measurement bars.
- **Canonical Circuit Presets:** 1-click loading for Bell State, GHZ State, Quantum Superposition, Teleportation, and Superdense Coding.
- **Dual Simulation Engine:** Connects to real Qiskit / NumPy engine (FastAPI backend) with automatic client-side matrix evolution fallback.
- **Multi-Dimensional Analytics:** Visualizes statevectors, measurement histograms across configurable shots (100–8192), and 3D Bloch spheres.
- **AI Circuit Explainer & Code Export:** Generates plain-English AI explanations of built circuits and exports to OpenQASM 2.0, Qiskit Python, SVG, and JSON.

4. Algorithm Lab (/algorithms)

Primary Purpose: Comprehensive theoretical and mathematical deep-dive into canonical quantum algorithms and computational speedups.

Key Features & Functions:

- **Curated Quantum Algorithms:** In-depth modules for Deutsch-Jozsa, Bernstein-Vazirani, Simon's Algorithm, Grover's Search, and Quantum Phase Estimation (QPE).
- **Structured Framework:** Every algorithm is broken down by Problem Statement, Classical vs. Quantum Complexity, Mathematical Derivation, and Step-by-Step Circuit Workflow.
- **Direct Lab Integration:** 'Load into Circuit Builder' launches the algorithm's circuit directly into the Quantum Lab for hands-on simulation.

5. Algorithm Playground (/playground)

Primary Purpose: Step-by-step, animated execution of quantum algorithms with dynamic 3D statevector and probability bar transformations.

Key Features & Functions:

- **Interactive Parameter Configuration:** Allows users to configure target marked items (for Grover) or oracle functions (constant vs. balanced for Deutsch-Jozsa).
- **Stage-by-Stage Stepper:** Step forward and backward through algorithmic stages (Initialization → Superposition → Oracle Evaluation → Amplitude Amplification → Measurement).
- **Dynamic 3D Visual Feedback:** Synchronized 3D probability bar charts and statevector trajectories reveal quantum interference in real time.

6. AI Quantum Tutor (/tutor (and Global Floating Widget))

Primary Purpose: 24/7 intelligent quantum mentor providing personalized, contextual explanations, step-by-step problem solving, and math derivations.

Key Features & Functions:

- **Conversational Intelligence:** Powered by Google Gemini with formatted mathematical equations (LaTeX), step breakdowns, and curated context.
- **Bring-Your-Own-Key Support:** Built-in API Key configuration modal allows users to connect their personal Gemini API key seamlessly.
- **Offline/Fallback Semantic Catalog:** 30+ pre-indexed quantum topics with instant search and intelligent query matching when offline.
- **Follow-Up Exploration:** Contextually suggests next-step questions and exploration paths based on user dialogue.

7. Practice & Challenges (/practice)

Primary Purpose: Interactive assessment arena to validate conceptual understanding and quantum circuit construction skills.

Key Features & Functions:

- **Categorized Concept Quizzes:** Multiple-choice question sets spanning Beginner, Intermediate, and Advanced tiers with detailed answer rationales.
- **Interactive Circuit Puzzles:** Problem challenges requiring learners to construct circuits that achieve target quantum states (e.g. generating entanglement or phase shifts).
- **Mastery & Score Tracking:** Awards XP, tracks accuracy rates, and logs milestone achievements on user profiles.

8. Dashboard (/dashboard)

Primary Purpose: Centralized personal dashboard providing learning analytics, recent experiments, and quick-access platform shortcuts.

Key Features & Functions:

- **Learning Analytics:** Visual metrics tracking completed Basics modules, simulated circuits count, and quiz mastery percentage.
- **Saved Circuits Repository:** Quick launch and management for saved custom quantum circuits.
- **Daily Recommendations:** Curated suggestions for next modules, trending algorithms, or daily quantum puzzles.

Platform Sections Summary Matrix

Quick-reference capability and technology overview across all modules.

Section	Route	Core Technology	Primary User Action
Landing Page	/	React, Three.js, Canvas WebGL	Try live 3D Bloch sphere & search platform
Quantum Basics	/basics	React, SVG, Interactive State	Learn foundational concepts step-by-step
Quantum Lab	/lab	Qiskit, NumPy, Three.js, QASM	Build, simulate & export quantum circuits
Algorithm Lab	/algorithms	React, LaTeX / KaTeX	Study math derivations & complexity theory
Playground	/playground	Three.js 3D, Step Execution	Inspect step-by-step algorithm state evolution
AI Tutor	/tutor	Google Gemini API, Local DB	Ask quantum questions with 24/7 AI mentor
Practice	/practice	State Evaluation, Quiz Engine	Solve circuit puzzles & test knowledge
Dashboard	/dashboard	LocalStorage / Supabase Auth	Track progress, stats & load saved circuits